Exam 14 June 2023

①. a) $A \in \mathcal{M}_n(\mathbb{R})$, $\eta \in \mathbb{R}^n$. Representation formula for the sol. of the IVP:

$x' = AX$, $X(0) = \eta$ → first-order lin. homog. system w c.c.

(THEORY) the unique sol: $X(t) = e^{t\cdot A}\cdot \eta$

$$e^{tA} = \sum_{k=0}^{\infty} \frac{t^k \cdot A^k}{k!}$$

$$X(t) = \left(\sum_{k=0}^{\infty} \frac{t^k A^k}{k!}\right)\eta \quad \text{repres. formula.}$$

b) $t \in \mathbb{R}$. Def. of the matrix exponential: $\boxed{e^{tA} = I + t\cdot A + \frac{t^2\cdot A^2}{2!} + \dots}$

• $e^{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}} = ?$

$A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$; $A^1 = A = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$A^2 = A = A^3 \dots$

$e^{tA} = e^{\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}} = I + 0 + 0 + \dots = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

• $e^{t\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}$

$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$; $A^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$A^3 = A^2 \cdot A = 0 = A^4 = \dots$

$$e^{tA} = I + t\cdot A + \frac{t^2\cdot A^2}{2!} + \dots$$
$$= I + \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix} + 0 \dots$$
$$= \begin{pmatrix} 1 & t+1 \\ 0 & 1 \end{pmatrix}$$

• $e^{t\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$

$A = I = A^2 = A^3 = \dots$

$$e^{tA} = I + t\cdot I + \frac{t^2\cdot I}{2!} + \dots = \Big(\underbrace{1 + t + \frac{t^2}{2!} + \dots}_{e^t}\Big) I$$

• $e^{t\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}$

$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; $A^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I$ $\Rightarrow I, A, -I, -A, I, A \dots$

$A^3 = -I\cdot A = -A$; $A^4 = -A\cdot A = -A^2 = I$

$$e^{tA} = I + t\cdot A - \frac{t^2}{2!} I - \frac{t^3}{3!} A + \frac{t^4}{4!}\cdot I \;\dots$$

$$= \underbrace{\left(1 - \frac{t^2}{2!} + \frac{t^4}{4!} \dots\right)}_{\cos t} I + \underbrace{\left(t - \frac{t^3}{3!} \dots\right)}_{\sin t} A$$

(Taylor series)

$$= (\cos t)\cdot I + (\sin t)\cdot A$$

$$= \begin{pmatrix} \cos t & 0 \\ 0 & \cos t \end{pmatrix} + \begin{pmatrix} 0 & -\sin t \\ \sin t & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

c) $A, J, P \in M_n(\mathbb{R})$

Assume P invertible and $A = P\cdot J\cdot P^{-1}$

$e^A = P\cdot e^J\cdot P^{-1}$?

$$A^2 = (PJP^{-1})(PJP^{-1}) = P\cdot J\cdot J\cdot P^{-1} = P\cdot J^2\cdot P^{-1}$$

$$P^{-1}P = I$$

$$A^3 = PJ^3P^{-1}$$

$$\vdots$$

$$A^k = PJ^kP^{-1}$$

$$\underset{(t=1)}{e^A} = I + t\cdot A + \frac{t^2\cdot A^2}{2!} + \dots$$

$$= I + 1\cdot A + \frac{A^2}{2!} + \dots$$

$$= I + PJP^{-1} + \frac{PJ^2P^{-1}}{2!} + \frac{PJ^3P^{-1}}{3!} + \dots$$

$$= P\cdot I\cdot P^{-1} + PJP^{-1} + \frac{PJ^2P^{-1}}{2!} + \frac{PJ^3P^{-1}}{3!} + \dots$$

$$= P\underbrace{\left(I + J + \frac{J^2}{2!} + \frac{J^3}{3!} + \dots\right)}_{e^J}P^{-1}$$

$$= P\cdot e^J\cdot P^{-1}$$

) Lotka-Volterra; planar sys.

$$\begin{cases} \dot{x} = x(1-y) \\ \dot{y} = y(2-x) \end{cases}$$

a) F.I. $(0,\infty) \times (0,\infty)$ → check.

$$\frac{dy}{dx} = \frac{y(2-x)}{x(1-y)} \Rightarrow x(1-y)\,dy = y(2-x)\,dx \Rightarrow$$

$$\Rightarrow (1-y)\,dy = y \cdot \frac{2-x}{x}\,dx \;\Big|\,:y \Rightarrow \frac{1-y}{y}\,dy = \frac{2-x}{x}\,dx \;\Big|\int$$

$$\Rightarrow \int\left(\frac{1}{y}-1\right)dy = \int\left(\frac{2}{x}-1\right)dx \Rightarrow$$

$$\Rightarrow \ln|y| - y = 2\cdot\ln|x| - x + C$$

$$\Rightarrow \underbrace{x - y + \ln|y| - 2\ln|x|}_{H(x,y),\ \cancel{H:\mathbb{R}\to\mathbb{R}}} = C$$

$H: (0,\infty)\times(0,\infty) \to \mathbb{R}$.

$$\frac{\partial H}{\partial x}(x,y)(x(1-y)) + \frac{\partial H}{\partial y}(x,y)(y(2-x)) \overset{?}{=} 0$$

$$\Leftrightarrow \left(1-\frac{2}{x}\right)x(1-y) + \left(-1+\frac{1}{y}\right)y(2-x) \overset{?}{=} 0$$

$$\Leftrightarrow (x-2)(1-y) + (1-y)(2-x) = 0$$

$$\Rightarrow (1-y)(x-2+2-x) = 0 \quad \text{true.} \quad \Rightarrow H \text{ global f.i.}$$

b) $(2,1)$ equil?

$\nabla H(x,y) = \left(1-\frac{2}{x}, -1+\frac{1}{y}\right) \Rightarrow H$ not locally ct.

$\nabla H(2,1) = (1-1, -1+1) = (0,0) \Rightarrow (2,1)$ indeed equil. ✓

$$f(x,y) = \begin{bmatrix} x-xy \\ 2y-xy \end{bmatrix}, \quad f(2,1) = \begin{bmatrix} 2-2 \\ 2-2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow (2,1) \text{ equil.} ✓$$

$$Jf(x,y) = \begin{bmatrix} 1-y & -x \\ -y & 2-x \end{bmatrix}$$

$$Jf(2,1) = \begin{bmatrix} -1 & -2 \\ -1 & 0 \end{bmatrix} \Rightarrow \begin{vmatrix} -1-\lambda & -2 \\ -1 & 0-\lambda \end{vmatrix} = 0$$

$$\Rightarrow -\lambda(-1-\lambda) - 2 = 0 \Rightarrow \lambda(1+\lambda) = 2 \Rightarrow \lambda^2+\lambda-2 = 0$$

$$\Delta = 1-4\cdot(-2) = 9 \Rightarrow \lambda_{1,2} = \frac{-1\pm 3}{2} \to \lambda_1 = 1, \ \lambda_2 = -2 \quad Re(\lambda) = 0 \Rightarrow \text{not hyp.}$$

(3) sol. of IVP:

a) $x' + 3x = 2, \quad x(0) = 0$

~~Int. factor:~~ $x_p = \frac{2}{3}$

~~$\mu(t) =$~~ $x' + 3x = 0$

$r + 3 = 0 \Rightarrow r = -3 \leftrightarrow c_1 \cdot e^{-3t}$

$x_h = c \cdot e^{-3t}$

$x(t) = c \cdot e^{-3t} + \frac{2}{3}$

$x(0) = c + \frac{2}{3} = 0 \Rightarrow c = -\frac{2}{3}$

b) $x_{k+1} + 3x_k = 2, \quad x_0 = 0$

~~$r + 3 = 2 \Rightarrow r = 1, c \in \mathbb{R}$~~

$\rightarrow \quad x_{k+1} + 3x_k = 0$

$r + 3 = 0 \Rightarrow r = -3 \longmapsto (-3)^k$

$x_h = x_k = c \cdot (-3)^k, \quad c \in \mathbb{R}.$

~~$x_p = \frac{2}{3}$~~

$x_k = a$ ct. $\Rightarrow x_{k+1} = a. = \frac{1}{2} = x_p.$

$\Rightarrow a + 3a = 2 \Rightarrow 4a = 2 \Rightarrow a = \frac{1}{2}$

$x = x_h + x_p = c \cdot (-3)^k + \frac{1}{2}$

$x_0 = 0 \Rightarrow c \cdot (-3)^0 + \frac{1}{2} = 0$

$\Rightarrow c + \frac{1}{2} = 0 \Rightarrow c = -\frac{1}{2}$

$x = x_k = -\frac{1}{2} \cdot (-3)^k + \frac{1}{2}$